Chapter 3

Data Communications and Networking

Data Communication Basics

- Data Communication is the exchange of data b/n devices via some form of transmission media (wired/wireless)
  - ✓ It needs communicating devices as hardware(physical equipment) and software (program)
- Effectiveness of data communication system depends on
  - ✓ Delivery. The system must deliver data to the correct destination. Data must be received by the intended device or user and only by that device or user.
  - ✓ Accuracy. The system must deliver the data accurately. Data that have been altered in transmission and left uncorrected are unusable.
  - ✓ Timeliness. The system must deliver data in a timely manner. Data delivered late are useless. In the case of video and audio

- ✓ Jitter refers to the variation in the packet arrival time. It is the uneven delay in the delivery of audio or video packets.

❑ Data Representation techniques

- ❑ The information can be different forms such as text, numbers, images, audio, and video.
- ❑ Text is represented as a bit pattern, a sequence of bits . Different sets of bit patterns have been designed to represent text symbols is coding (such as Unicode 32 bit ,ASCII)
- ❑ Numbers are represented by bit patterns which converted to binary number to simplify the mathematical operation.
- ❑ Images are also represented by bit patterns which is composed of a matrix of pixels where each pixel is a small dot.
- ❑ Audio refers to the recording or broadcasting of sound or music in the form of continuous electrical signal which is different from text, image, number and so on.

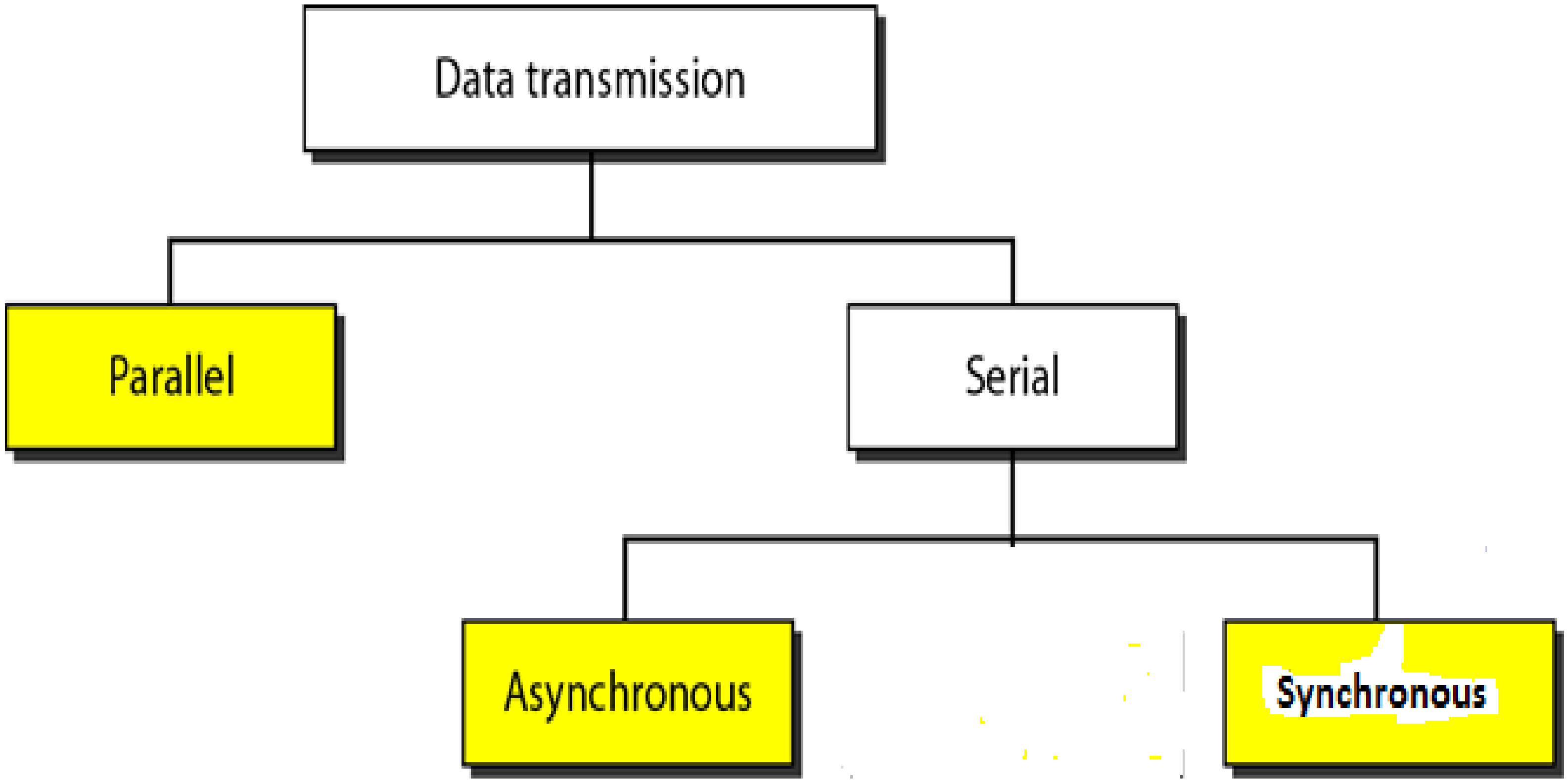
Cont'd …

- Video refers to the recording or broadcasting of a picture or movie. Video can either be produced as a continuous entity (e.g., by a TV camera) or discrete entity.
- **What is computer network**
- **Network** is a set of devices (often referred to as nodes) connected by communication links.
- The Internet evolved from the *ARPANET*, which was developed in 1969 by the Advanced Research Projects Agency (ARPA) of the U.S. Department of Defense.

Data Transmission

- It is sending of binary data from source to destination as bit by bit or grouping bits together,
- But how it could be?
- Ans:
- The transmission of binary data across a link (communication channel) can be either parallel or serial mode.
- This means?
- In parallel mode multiple bits are sent together each clock, where as
- In serial mode, 1 bit is sent with each clock tick.

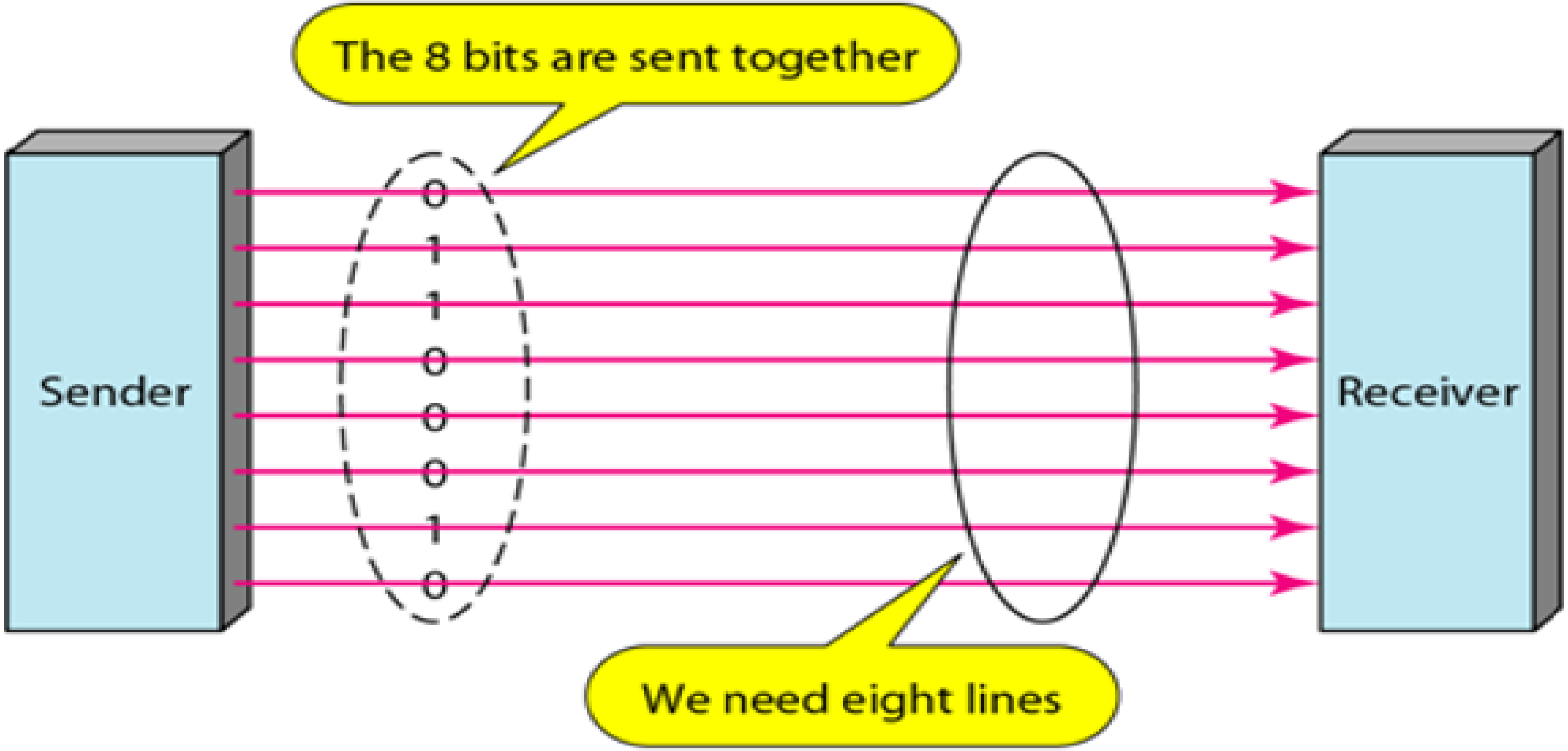
Types/ways of Data Transmission



Parallel Data Transmission

- It organized bits into groups of *n* bits each to send.
- Computers produce and consume data in groups of bits as human spoken language use in the form of words rather than letters.
- It sends *data* in *n bits at a time instead of 1*.
- It is a conceptually simple mechanism than serial
- Use *n* wires to send *n* bits at one time.
- each bit has its own wire, and all *n* bits of one group can transmit
- Its *advantage* over *serial transmission* could be high *speed*.

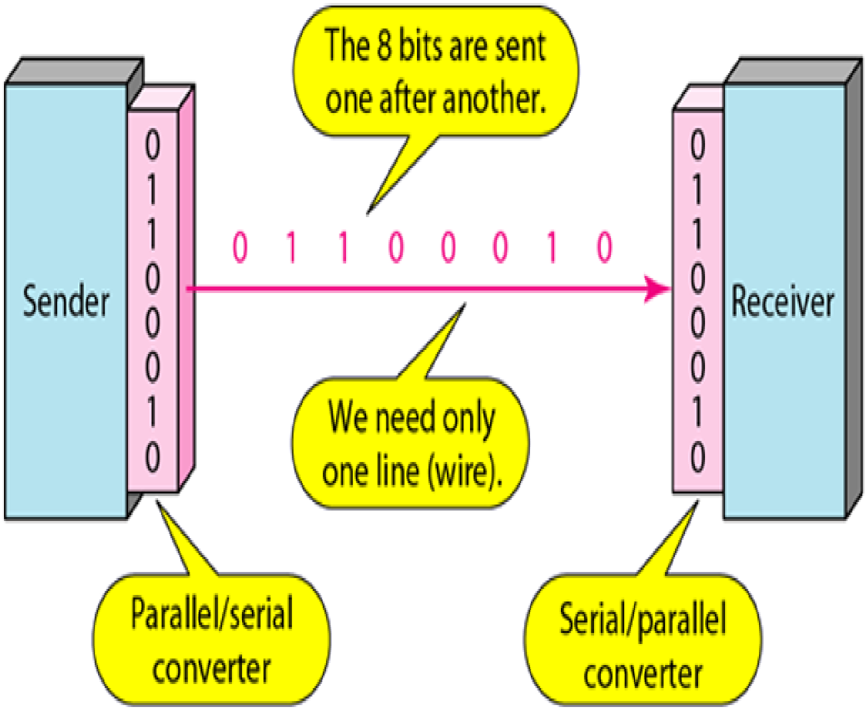
Parallel Data Transmission…

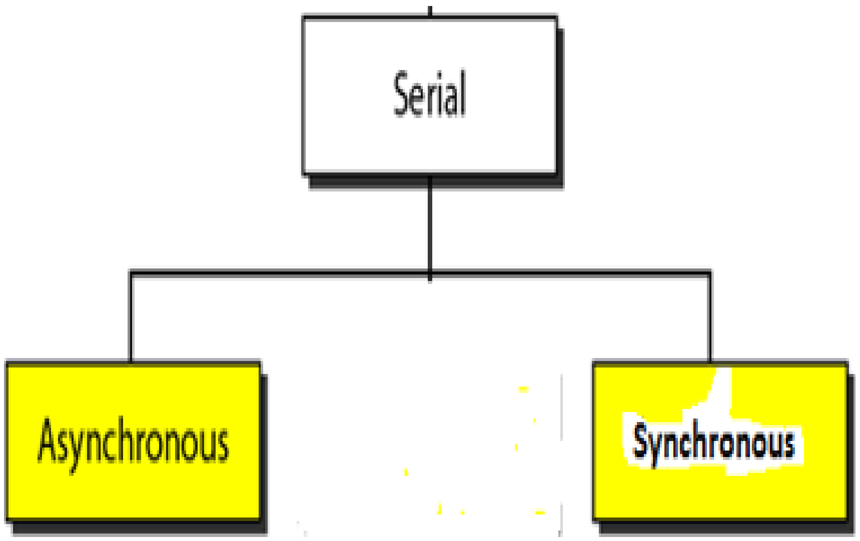


Serial Data transmission

- Signals are sent one bit at a time
- Travels long distances
- Serial transmission reduces the cost of transmission over parallel by roughly a factor of *n*.
- Example: telephone wires

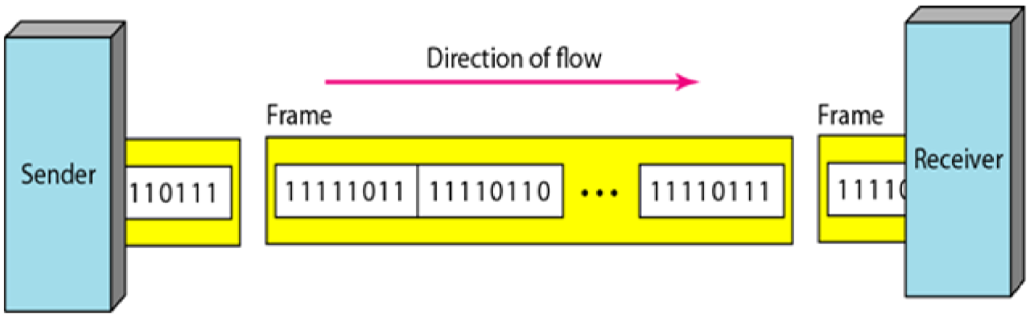
Serial Data transmission…





1. Synchronous data transmission…

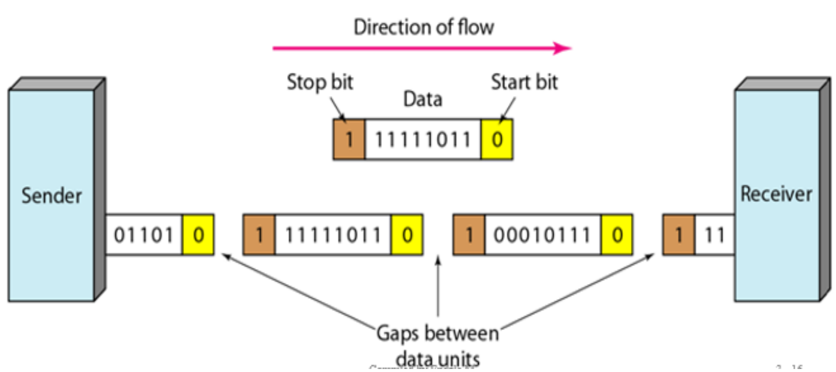
- Receiver waits ready for sender message and responds in real time (e.g. phone call).



- High speed.
- So, useful for high-speed applications such as the transmission.
- Does not use stop/start bits – instead devices agree on timing
- No overhead of bits b/c start and stop bits
- No buffer is required.
- Examples: phone call

2. Asynchronous data transmission

- Both sender and receiver no required clock signals
- It sends data by grouping it as bytes
- Have added parity bits (which called start and stop bit) for synchronize clock signals b/n sender and receiver.
- There is gap b/n frames or group of bits (bytes) by start and stop bits
- Need buffer for data until synchronize both sender and receiver
- (e.g. mailbox).



Signals

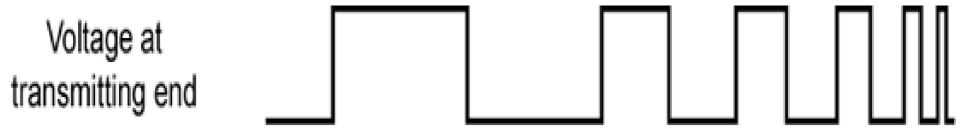
❑It is propagation of data from one point to another by means of electromagnetic signals.

❑It can be either analog or digital signal

❑A **digital signal** is one in which the signal intensity maintains a *constant level for some period of time*

❑It have only a limited number of defined values

❑ It can be represented by digital signals, with a different voltage level for each of the two binary digits.



❑Used to represent *digital data*

Digital Signals

Advantages of Digital Signals

- ❑cheaper than analogy signalling
- ❑Less susceptible to noise

Disadvantages of Digital Signals

- ❑Suffer more from attenuation!
  - ❖Pulses become rounded and smaller
  - ❖Leads to loss of information

2. Analog Signals

- **Analog signal** is the simplest sort of signal is a in which the same signal pattern repeats over time and
- It can have an infinite number of value in a range
- Used to represent *analog data*
- *Analog data – information that is continuous and take continuous values*

2. Analog Signals

- **Analog data** are a function of time and occupy a limited frequency spectrum; such data can be represented by an electromagnetic signal occupying the same spectrum.
- Can use analog signal to carry digital data by using Modem:
- The modem converts a series of binary voltage pulses into an analog signal by encoding the digital data onto a carrier frequency.
- Can use digital signal to carry analog data with codec which takes an analog signal and approximates that signal by a bit stream.

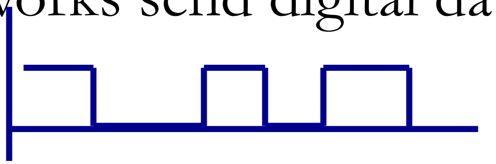
I. Digital Data Transmission

1. Digital data

- Produced by computers, in binary form, represented as a series of ones and zeros
- Can take on only **0** and **1**

2. Digital transmissions

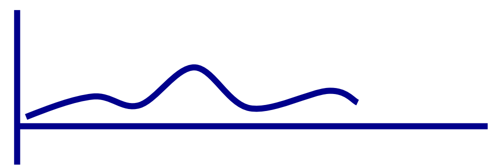
- Made of square waves with a clear beginning and ending
- Computer networks send digital data using digital transmissions.



II. Analog Data Transmission

1. Analog data
   - ❖Produced by telephones
   - ❖Sound waves, which vary continuously over time
   - ❖Can take on any value in a wide range of possibilities

2. Analog transmissions



   - ❖Analog data transmitted in analog form (vary continuously)
   - ❖Examples of analog data being sent using analog transmissions are broadcast TV and radio

❑Data converted between analog and digital formats
   - ❖Modem (modulator/demodulator): used when digital data is sent as an analog transmission
   - ❖Codec (coder/decoder): used when analog data is sent as a digital transmission

Data Transmission Mode

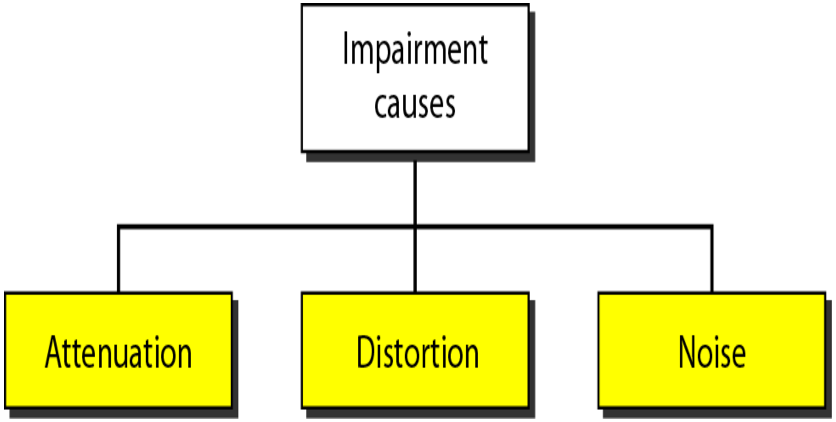
- Simplex transmission
  - Signals are transmitted in only one direction
    - e.g. Television
- Half duplex
  - Signals can be transmitted in either direction, but only one way at a time.
    - e.g. police radio
- Full duplex
  - Both stations may transmit simultaneously.
    - e.g. telephone

Transmission Impairments and their solution

- Signals travel through transmission media, which are some time ***not perfect.***
- The imperfection causes signal impairment.
- This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium.
- What is sent is not what is received.
- With any communications system, the signal that is received may differ from the signal that is transmitted, due to transmission impairments

Transmission Impairments and their solution…

- Consequences over two types of signals we have:
  - For analog signals: degradation of signal quality
  - For digital signals: bit errors
- The most significant impairments include
  - **Attenuation**
  - **distortion**
  - **Noise**



Transmission Impairments and their solution…

1. Attenuation

- Attenuation means a loss of energy.
- When a signal, simple or composite, travels through a medium, it loses some of its energy in overcoming the resistance of the medium.
- That is why a wire carrying electric signals gets warm, if not hot, after a while.
- Some of the electrical energy in the signal is converted to heat.
- This conversion of electrical energy causes loos of signal strength (attenuation)

Transmission Impairments and their solution…

Measurement of Attenuation

- To show the loss or gain of energy the unit "decibel" is used.

$$dB = 10\log_{10} P_2/P_1$$

Where as

P_1 is input signal and P_2 is output signal

- These problems are dealt with by the use of amplifiers or repeaters by amplifying the signal..

Transmission Impairments and their solution…

Example 1 of Attenuation…

- Suppose a signal travels through a transmission medium and its power is reduced to one-half. This means that P2 is (1/2)P1.
- In this case, the attenuation (loss of power) can be calculated as

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{0.5 P_1}{P_1} = 10 \log_{10} 0.5 = 10(-0.3) = -3 \text{ dB}$$

- A loss of 3 dB (–3 dB) is equivalent to losing one-half the power.

Transmission Impairments and their solution…

Example 2 of Attenuation…

- A signal travels through an amplifier, and its power is increased 10 times. This means that $P_2 = 10P_1$.
- The compute the amplification or gain of power by amplifier.

Solution:

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{10P_1}{P_1} = 10 \log_{10} 10 = 10(1) = 10 \text{ dB}$$

Transmission Impairments and their solution…

2. Distortion

- Means that the signal changes its form or shape.
- Distortion can occur in a composite signal made of different frequencies
- Each frequency component has its own propagation speed traveling through a medium.
- The different components therefore arrive with different delays at the receiver.
- That means that the signals have different phases at the receiver than they did at the source.

Transmission Impairments and their solution…

3. Noise

- Several types of noise, such as
- **Thermal noise** is the random motion of electrons in a wire which creates an extra signal not originally sent by the transmitter.
- **Induced noise** comes from sources such as motors and appliances that act as antenna and medium as receiving antenna.
- **Crosstalk** is the effect of one wire on the other. It is an unwanted coupling between signal paths and can occur by electrical coupling between nearby twisted pairs.
- **Impulse noise** is a spike (a signal with high energy in a very short time) that comes from power lines, lightning, and so on

Transmission Impairments and their solution…

3. Noise

- Noise can be
  - Electromagnetic Interference
  - Radio frequency interference

- noise can be reduce by

- Twisting cables – effect of one signal cancels the other
- Shielding – reduce interference from outside source

Transmission Impairments and their solution…

Signal to Noise Ratio (SNR)

- To measure the quality of a system the SNR is often used.

- It is the ratio between two powers (input and output).
  - It is usually given in dB and referred to as SNR_{dB}.

$$SNR = \frac{\text{average signal power}}{\text{average noise power}}$$

Example:

The power of a signal is 10mW and the power of the noise is 1μW; what are the values of SNR and SNR_{dB} ?

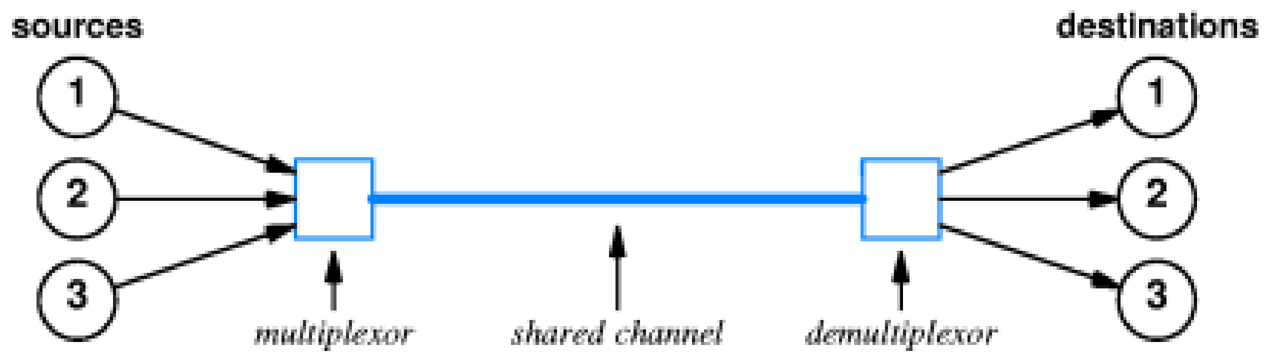
Solution

$$SNR = \frac{10{,}000\ \mu W}{1\ mW} = 10{,}000$$

$$SNR_{dB} = 10 \log_{10} 10{,}000 = 10 \log_{10} 10^4 = 40$$

Multiplexing/Demultiplexing

- Multiplexing/Demultiplexing: Used when many source communicate with many destinations through one communication line.



Types of Multiplexing